Day 5 — Git & GitHub

Complete answers and practical guide

Task 1: Introduction to Git and GitHub

Git: Git is a version control system used to track changes in files and manage different versions of a project.

GitHub: GitHub is an online platform used to store Git repositories, share code, and collaborate with other developers.

Difference: Git tracks changes on your computer; GitHub hosts repositories online. For example, Git tracks changes in a Java project, while GitHub lets you store and share that project online.

Task 2: Create a GitHub Account

1. Open https://github.com.

2. Click **Sign up**.

3. Enter your email address and create a password.

4. Choose a username and complete verification.

5. Verify your email address.

Result: Your GitHub account is ready. If you already have an account, skip this task.

Task 3: Create a Repository for Internship Tasks

A **repository (repo)** is a place where project files and their version history are stored.

1. Log in to GitHub and click the **+** icon in the top-right corner.

2. Select **New repository**.

3. Enter repository name: `internship-tasks`.

4. Add description: “Programs completed during internship training.”

5. Select Public or Private.

6. Optionally select **Add a README file**, then click **Create repository**.

Result: A repository named `internship-tasks` is created.

Task 4: Install and Configure Git

Step 1: Install Git

Visit https://git-scm.com/downloads, download Git for your operating system, install it, then open Git Bash or Command Prompt.

```
git --version
```

Example output: `git version 2.x.x` (the exact version depends on your installation).

Step 2: Configure Git

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
git config --global --list
```

Replace the example name and email with your own. Git uses these details to identify the author of your commits.

Task 5: Upload Days 1–4 Programs to GitHub

Step 1: Organize your files

```
internship-tasks/
|-- Day1/   (program files)
|-- Day2/   (program files)
|-- Day3/   (program files)
|-- Day4/   (program files)
`-- README.md
```

Put your actual Day 1–4 programs in the appropriate folders. Open Git Bash or a terminal inside the project folder.

Step 2: Initialize Git and commit files

```
git init
git add .
git commit -m "Add Day 1-4 internship tasks"
git branch -M main
```

Step 3: Connect your GitHub repository and push

```
git remote add origin https://github.com/YOUR-USERNAME/internship-tasks.git
git push -u origin main
```

Replace `YOUR-USERNAME` with your GitHub username. Authenticate with GitHub if prompted.

Important: These commands assume the GitHub repository is empty. If you created it with a README, clone it first or integrate the remote changes before pushing.

Expected output

The GitHub repository should show your Day 1, Day 2, Day 3, and Day 4 folders and their program files.

Task 6: Practice Basic Git Commands

`git init` — Initializes a Git repository.

`git status` — Shows the current state of files.

`git add .` — Stages all eligible changes.

`git add filename` — Stages one specific file.

`git commit -m "message"` — Records staged changes in Git history.

`git log` — Displays commit history.

`git branch` — Lists local branches.

`git branch new-branch` — Creates a branch.

`git switch main` — Switches to the main branch.

`git clone URL` — Copies a remote repository to your computer.

`git push` — Uploads local commits to a remote repository.

`git pull` — Fetches and integrates remote changes.

`git diff` — Displays changes that have not been staged.

`git merge branch-name` — Merges the named branch into the current branch.

Example workflow

```
git status
git add .
git commit -m "Update internship programs"
git push
```

This checks the changes, stages them, records a commit, and uploads the commit to GitHub.

Task 7: Branching Basics

A **branch** is a separate line of development. It lets you work on changes without immediately modifying the main branch.

Create and switch to a branch

```
git branch feature
git switch feature
```

Or create and switch to the branch in one command:

```
git switch -c feature
```

Merge a branch

```
git switch main
git merge feature
```

This merges changes from `feature` into the current `main` branch. Test your changes before merging.

Task 8: Create a README File

A **README** explains a project's purpose, contents, and usage. GitHub displays `README.md` prominently on the repository page.

Create a file named `README.md` and add this example, editing it to match your actual tasks:

```
# Internship Tasks

## Introduction
This repository contains programs completed during my internship training.

## Tasks
### Day 1
Programs and exercises completed on Day 1.
### Day 2
Programs and exercises completed on Day 2.
### Day 3
Programs and exercises completed on Day 3.
### Day 4
Programs and exercises completed on Day 4.

## Technologies Used
- Java (if applicable)
- Git
- GitHub

## Purpose
To practise programming concepts and maintain a record of my learning progress.

## Author
Your Name
```

Replace the descriptions with the actual topics and technologies used in your Day 1–4 tasks.

Task 9: Expected Output

1. A GitHub account is created, if required.

2. A repository named `internship-tasks` is created.

3. Day 1–4 programs are uploaded.

4. A README file explains the repository contents.

5. Basic Git commands are practised.

6. The uploaded files are visible on GitHub.

Final result: The internship task programs are organized and uploaded to GitHub, where they can be accessed and maintained online.

Task 10: How to Explain This Task in an Interview

“I created a GitHub repository to store my internship tasks. I uploaded my Day 1 to Day 4 programs and created a README file to explain the repository contents. I also practised basic Git commands such as `git init`, `git add`, `git commit`, `git push`, `git pull`, and `git clone`. This helped me understand how to manage code versions and store projects on GitHub.”

Short version: “I used Git for version control and GitHub to store my internship programs online. I created a repository, uploaded my Day 1–4 tasks, added a README file, and practised basic Git commands.”

Quick Revision: Important Git Terms

Git: A version control system.

GitHub: An online platform for hosting Git repositories.

Repository: A project folder tracked by Git.

Commit: A recorded set of changes.

Branch: A separate line of development.

Push: Upload local commits to a remote repository.

Pull: Fetch and integrate remote changes.

Clone: Copy a remote repository to your computer.

README: A file explaining a project.

Merge: Combine changes from one branch into another.

Completion checklist: Create the repository, upload the Day 1–4 programs, add the README, and practise the basic Git commands.